This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCAD.2017.2695881, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems



# A Genetic Algorithm based Heuristic Method for Test Set Generation in Reversible Circuits

Nagamani A N, *Member, IEEE*, Anuktha S Nayak, Nanditha N, Vinod Kumar Agrawal, *Senior Member, IEEE*

***Abstract*—Low power circuit design has been one of the major growing concerns in integrated circuit technology. Reversible circuit (RC) design is a promising future domain in computing which provides the benefit of less computational power. With the increase in the number of gates and input variables, the circuits become complex and the need for fault testing becomes crucial in ensuring high reliability of their operation. Various fault detection methods based on exhaustive test vector search approaches have been proposed in the literature. With increase in circuit complexity, a faster test generation method for providing optimal coverage becomes desirable. In this paper, a Genetic Algorithm (GA) based heuristic test set generation method for fault detection in reversible circuits is proposed which avoids the need for an exhaustive search. Two approaches, one involving random search and the other, involving directed search have been proposed and validated on benchmark circuits considering missing-gate fault (complete and partial), bridging fault and stuck-at fault with optimum coverage and reduced computational efforts.**

***Index Terms*—Reversible logic, Genetic Algorithm, ATPG, Fault Coverage, Computation Time**

## I. Introduction

With the rising demands for quantum computing methods for their promising low power dissipation properties and higher data representation and retention capabilities, the field has witnessed rapid advancements. Landauer's principle [1] states that logical reversibility is associated with physical reversibility which serves the purpose of minimal heat generation per machine cycle. The device is said to be logically irreversible if its output is not uniquely defined by the inputs. A logic circuit is reversible if it computes bijective (one-to-one and onto) logic functions. No information loss occurs during the entire operation, thus lowering power consumption to a large extent. Bennet [2] showed that, if computations are carried out irreversibly, $kT \ln 2$ (where $k$ is the Boltzmann constant- $1.3806 \times 10^{-23} m^2 kg^{-2} K^{-1}$ and $T$ is the absolute temperature at which computation takes place) power dissipation would occur for each bit of information erasure or throwaway. Hence, lossless computation must be reversible. Reversible circuits used with newer technologies such as quantum computing [3], optical computing [4], Quantum Cellular Automata (QCA) [5], trapped-ion technology [6], adiabatic CMOS [7] offer reduction in power dissipation. Different types of reversible gates including but not limited to Toffoli [8], Fredkin [9] and Peres [10] have been logically designed and many reversible logic circuits are being built with the same. As the versatility of reversible circuits increase with its capability to handle a large number of gates and its bits storage capacity, the vulnerability of the circuits to potential faults increases and thus the requirement for a testing system to detect faults in such circuits becomes very crucial to ensure the correctness of the implemented logic circuit.

Authors in [11], [12] and [13] have proposed various testing algorithms for reversible gates using SAT and Exact methods, respectively. As described in these papers, one common approach is to test the circuit using the entire set or at least a majority of the $2^n$ test vectors ($n$ being the number of lines in the RC) which makes the algorithm exponentially complex for $n \geq 15$. Since the test generation process is an NP-hard problem and exponentially complex, an exhaustive search approach quickly becomes impractical as the search space grows colossal in size. GA thus provides a heuristic approach for arriving at the optimum solution by attempting to find better solutions in each iteration until a threshold criterion is met. GA based algorithms have been extensively studied and found to be very useful in solving NP-hard problems [14], where the problems cannot be characterized by a polynomial expression. An improvisation over GA by taking into account the quality of individuals (test vectors) selected in each generation– an adaptive algorithm [15] can be developed to obtain an optimum solution in a reasonable amount of time.

In this paper, an alternate solution is proposed to the existing testing methods which is efficient in terms of time, space and effort required to obtain a test set, and which can detect a large number of faults for small ($n<8$), medium ($8<n<30$) and large ($n>30$) circuits,

Nagamani A N, Anuktha S Nayak and Nanditha N are with the Department of Electronics and Communication Engineering, PES Institute of Technology, Bangalore, India (e-mail:nagamani@pes.edu).

Vinod Kumar Agrawal is with the Department of Information Science and Engineering, PES Institute of Technology, Bangalore, India

0278-0070 (c) 2016 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCAD.2017.2695881, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems



with ranges as found in [16]. Quantum circuits implemented on QCA tend to have defects such as cell displacement fault and cell misalignment faults which can be tested with stuck-at fault models [17]. Authors in [18] have discussed the applicability of various Register Transfer Level fault models such as single and multiple stuck-at faults that deal with Quantum gate interconnection which can be readily adapted to Quantum Circuits. According to [19] and [7], classical fault models−stuck-at faults, delay faults, missing-gate faults, wrong gate faults and bridging faults−need to be considered for RCs implemented as information reversible circuits and, missing-gate faults and wrong gate faults for RCs implemented as energy reversible circuits.

The proposed work focuses on the set of all missing-gate faults which can occur in both information and energy reversible circuits, stuck-at faults and bridging gate faults which find their relevance in information reversible circuits only. Using GA for fault detection eliminates the requirement of testing with all the $2^n$ vectors thereby reducing the overall computation time. The proposed GA based testing with a Random search approach and a Directed search approach have been validated on small, medium and large benchmark circuits from the libraries−Negative control Toffoli (NCT), Multiple control Toffoli (MCT), Multiple control Fredkin (MCF) and Peres (P) as referred in [20] and [21]. The algorithm has been coded using C language and applied to circuits in .tfc and .real formats on a computer with an Intel i5 processor operating on 4 GB RAM at 1.7 GHz clock speed.

The major contributions of this work are:

1 A heuristic GA based strategy using Random search method for obtaining optimum fault coverage for RCs designed using Toffoli, Fredkin and Peres gates for faults- single stuck-at, bridging and set of all missing-gate faults (partial and complete missing-gate faults).
2 Directed search approach, an improvisation over Random search approach is proposed which reduces the computation time significantly for $k$-CNOT circuits.
3 A single test set to detect a combination of faults for the set of all missing-gate faults.

The rest of the paper is organized as follows:
Section II briefs about existing reversible circuit testing algorithms. In Section III, various fault models and their detection methodologies relevant for reversible circuits are discussed and a brief on GA is given. Section IV presents the proposed algorithm with an illustrative example. Section V discusses the results obtained and compares it with recent work in this domain. Section VI concludes the work and scope for future work follows.

## II. Literature review on reversible circuits testing

Various fault detection methodologies are described in the literature for the testing of RCs. There are two major classifications of testing, namely, online and offline testing. Online testing is a process that can be carried out during normal operation whereas, offline testing is performed on the circuit in the test mode. Various online testing methods applied for RCs using dual-rail coding, NAND blocks, design for testability (DFT) methods are discussed in [22], [23], [24], [25] and [26].

An offline testing methodology using Integer Linear Programming (ILP) is proposed in [27]. Various fault models affecting RCs have been considered and tested for in the literature. A detection methodology for missing control fault model is proposed in [28], which includes several subsets of this model−single missing-gate fault (SMGF) also termed as complete missing-gate fault (CMGF), multiple missing-gate fault (MMGF), repeated gate fault (RGF) and partial missing-gate fault (PMGF)−emphasizing that, modelling of reversible circuit faults using only single stuck-at fault would not suffice. Authors in [19] state that only missing-gate faults are possible fault models to be considered in quantum reversible circuits. Detection of multiple missing-gate faults and repeated missing-gate faults have been carried out in [29], [30], [31] and [32], and a test set for combined fault models has been derived. Work in [33] indicates that cross point models are better suited for reversible circuits than stuck-at fault models. Bridging faults are more frequent as compared to stuck-at and missing-gate faults and are complicated to detect. A universal test set for bridging faults has been proposed in [34]. Further, an exact approach for bridging fault detection with a minimal test set is applied to reversible circuits in [35]. A Boolean satisfiability based fault detection for missing-gate faults is dealt with in [11]. Work in [36] proposes a method for the fault detection of Additional control and Repeated gate faults. Recent work in [37] proposes that symmetric functions with only PMGF can be detected with a maximum of 3 vectors (for any $n$).

Genetic Algorithm has been explored in literature for reversible and irreversible circuit synthesis and optimisation. A design based on GA for space exploration framework for parameterized System-On-Chip (SOC) platforms is proposed in [38]. Fault detection in irreversible circuits using GA has been proposed in [39] for the testing of Single Event Upsets in SRAM-FPGA, in [40] for testing large synchronous circuits and [41] employs GA to obtain an Automatic Test Vector Cultivation for combinational and sequential VLSI circuits. However, GA for fault detection and test set generation in reversible circuits has not yet been explored. With faults

0278-0070 (c) 2016 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCAD.2017.2695881, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems



in quantum circuits following a probabilistic distribution, stochastic methods for detection and diagnosis would be very effective [42]. GA being a heuristic approach can therefore be expected to give favorable results for testing RCs which find their applicability in quantum computing.

## III. Preliminaries

### A. Reversible Logic, Fault Models and detection methodology

RCs, unlike irreversible circuits have the property of input retrieval through output as every input combination has a unique output combination. Standard reversible gates used to build RCs are shown in Fig. 1, with their logical expressions in Table I. Quantum Cost (QC), Gate Count (GC), Ancilla Inputs (AI), Garbage Outputs (GO) and Delay ($\triangle$) govern the performance of these circuits. Toffoli and Feynman gates come under the class of $k$-CNOT gates with k values 3 and 2 respectively. The $k$-CNOT gate is a k-input, k-output reversible gate which transforms $[V_1, V_2, ...V_{k-1}, V_k]$ to $[V_1, V_2, ..., V_{k-1}, (V_1 \cdot V_2 \cdot ... \cdot V_{k-1}) \oplus V_k]$. The functionality of a Peres gate is the same as that of a cascade of Toffoli and Feynman gates. Fredkin gate functions as a controlled swap with input A being the control. A class of Fredkin gates with multiple controls are classified under MCF.

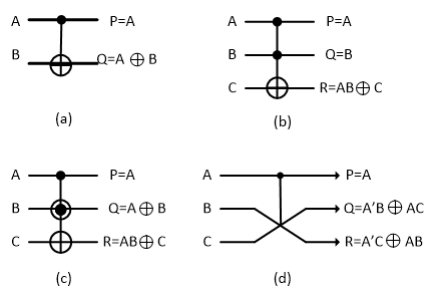


Fig. 1: (a).Feynman(F), (b).Toffoli(T), (c).Peres (P), (d). Fredkin (FR)

TABLE I: Reversible gates and Parameters

| Gate | function | QC |
|---|---|---|
| Feynman(F) | $F(A, B) = (A, A \oplus B)$ | 1 |
| Toffoli(T) | $F(A, B, C) = (A, B, AB \oplus C)$ | 5 |
| Peres (P) | $F(A, B, C) = (A, A \oplus B, AB \oplus C)$ | 4 |
| Fredkin (FR) | $F(A, B, C) = (A, AB' + AC, AB + A'C)$ | 5 |

Similar to irreversible logic circuits, reversible circuits have a set of fault models described to categorize the various faults that can be observed in them [43]. The ATPG algorithm presented in Section IV is applied on RCs for the detection of faults described here by inducing them at each potential fault location. For every fault induced, the faulty output of the circuit is compared with the fault-free output $R$ for a test vector chosen as input. If the two outputs differ, the vector is said to detect the fault and variable $detected$ is incremented by 1 from an initial value of 0.

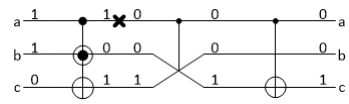

Fig. 2: Illustration of stuck-at 0 at the input of level 1

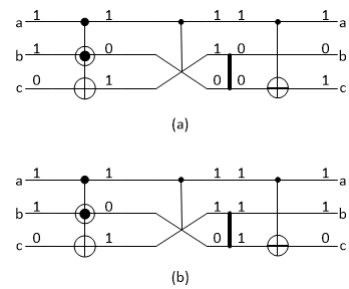


Fig. 3: (a) Illustration of 'b' AND bridged with 'c' at the input of level 2, (b) Illustration of 'b' OR bridged with 'c' at the input of level 2

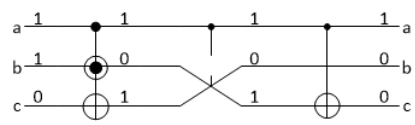

Fig. 4: Illustration of PMGF with a missing control for Fredkin

**1.** Stuck-at fault (SaF), Fig. 2: This fault model describes a line in the circuit at any level, stuck-at a particular signal value (0 or 1), denoted as x in Fig. 2, and does not allow any other signal value during circuit operation to change it. The majority of the faults in circuits can be mapped to this type of fault. Such a fault usually occurs during the manufacturing process. Faults of this kind are induced by forcing the input lines to 0 and 1, one at a time for all the levels of the RC with outputs computed in each case which is then compared with $R$, the fault-free output for the same test vector. $detected$ is incremented in case of a mismatch. The number of stuck-at faults possible in an $n$ input RC with $N$ gates are,

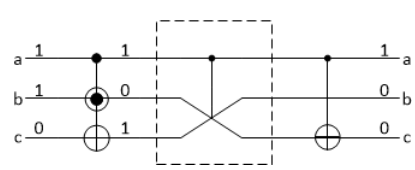

Fig. 5: Illustration of CMGF with a missing Fredkin gate

0278-0070 (c) 2016 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCAD.2017.2695881, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems



$$CumulatedFault = 2.n.(N+1)$$

**2.** Bridging fault (BF), Fig. 3: When the signal value on a line is influenced by the signal value on one of the other lines in the circuit, the line under consideration is said to be bridged. Depending on the way in which a line influences the signal value on the bridged line, these faults are classified into two types:
AND bridging: If line 'a' is AND bridged with line 'c', upon induction of the fault, 'a' and 'c' get the logical value 'a AND c'.
OR bridging: If line 'a' is OR bridged with line 'c', upon induction of the fault, 'a' and 'c' get the logical value 'a OR c'.
BF is introduced by picking two lines at a time from the $n$ lines of the RC. The inputs to these lines are AND*ed* and the common output is forced through the lines. Every possible combination of two lines is made starting from the inputs to the first gate to the output lines after the last gate. In the case of a mismatch, $detected$ is incremented. The entire process is repeated by OR*ing* the input lines. Total number of such intra-level faults are,

$$CumulatedFault = \binom{n}{2}.(N+1).2$$

**3.** missing-gate fault: Due to manufacturing defects, it might be possible for a gate in the circuit to be skipped or one or many of the control lines to a gate in the circuit go amiss; faults of this kind are labeled missing-gate faults. Two types of missing-gate faults are considered for testing in this work:
PMGF, Fig. 4: One or more of the control lines in a gate of an RC are missing giving rise to PMGF. The $CumulatedFault$ with one missing control is the sum total of all controls present in the gates of the RC. In order to calculate PMGF, the output $R$ is calculated for a test vector. The same test vector is then applied to the RC with controls skipped one at a time. The resulting output is compared with $R$ and $detected$ is incremented in case of a mismatch.
CMGF, Fig. 5: An entire level or gate is skipped from the circuit (reducing the gate count by 1) constituting a CMGF. A circuit can have as many CMGF as the total number of gates in it. In this case, $CumulatedFault$ is $N$. $detected$ for CMGF is calculated as in PMGF, with a gate skipped one at a time instead of the controls.
The same algorithm can also be extended to Repeated gate faults and Additional control faults. After obtaining $detected$ the fault coverage is calculated as,

$$FaultCoverage = \frac{detected}{CumulatedFault} \times 100 \qquad (1)$$

### B. Genetic Algorithm

Genetic Algorithm (GA) is a popular evolutionary algorithm developed based on the observation of the sustenance of genes from a population of living organisms. The odds of the survival of the genes depend on their fitness values. GA creates successive generations of individuals by applying simple reproduction operations. Typically, parents are chosen to mate with probability proportional to their fitness called proportional selection. This process of selection and reproduction continues until a maximum fitness is obtained. In circuit testing, the most commonly used operators include bit-string crossover, where, off-springs are formed by swapping a sub-sequence between the two parent bit strings, and bit-flipping mutation in which a single bit in the string is flipped to form a new offspring string [44]. Mutation helps in faster convergence and its probability of application decreases exponentially with generations [45] and [46].
Fault detection in reversible circuits is easy as they are bijective functions with higher controllability and observability [47]. This property enables detection of maximum faults with very few undetectable faults as most of the faults can be activated through controllable inputs and observed at primary outputs. At the same time, to ensure the property of reversibility, the number of input lines should be made equal to the number of output lines and with an increase in $n$, the number of potential fault sites increases drastically. An efficient ATPG must converge fast with maximum fault coverage consuming less memory space. ATPG using an exhaustive search as $n$ increases requires more effort in terms of computation time and memory to exercise the complete circuit for minimum test set generation.
Genetic algorithm, unlike other random sampling techniques, steers the search towards more prospective regions due to the combination of reproduction and mutation as directed by the fitness function. While gradient descent techniques can at best arrive at a local optimum, Genetic Algorithm can take values from various regions in the spread out curve of the function and hence can arrive at a far superior solution.
In fault detection using GA, test vectors are selected in a way so as to minimize the cost associated with it and previously selected vectors are also taken into consideration to select the filial generation. This ensures that the cost associated with the test set decreases with each generation until no further reduction is possible. This way, an optimum solution can be arrived at without sifting through the entire search space making GA the ideal solution to such problems which cannot be solved using other approaches.

## IV. Proposed ATPG algorithms

### A. GA incorporating Random search

In this section, GA along with its variant where the algorithm is directed by selecting the initial vector

0278-0070 (c) 2016 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCAD.2017.2695881, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems



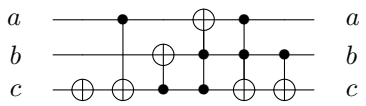


Fig. 6: 3_17$tc$ circuit

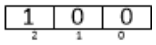


Fig. 7: Test vector as a chromosome

population for fault detection is presented. Algorithm 1 describes the basic structure of the GA for fault detection, incorporating a Random search approach to produce a test set for a given RC with $n$ input lines and $N$ gates. $Threshold$ indicates the desired percentage of fault coverage with the maximum being 100%. The next subsection expands the steps with the help of an example circuit.

The path taken by the GA is determined by the initial set of vectors chosen. Random selection does not always guarantee maximum coverage for every test run. It might take many iterations to yield optimum test set if the initially selected vectors are weak, which would lead to higher computation time or might even require early termination before optimum coverage has been reached. Thus, by directing the search, faster convergence would be highly likely, especially for MCT-based RCs with a large number of input lines.

### *B. Illustration of proposed algorithm using random search for Complete-missing-gate fault detection*

Consider the benchmark circuit 3_17$tc$ as shown in Fig. 6 as the input for the proposed algorithm to find a test set for optimum coverage of CMGF. In this algorithm, gates are also referred to as levels and $V_i$ in a set represents the i$^{th}$ element of the set.

**Illustration 1** Illustration for GA using Random search for the circuit 3_17$tc$

**INPUT:** RC with 'n' lines and 'N' gates; $Threshold$
**OUTPUT:** The test set and fault coverage%

**STAGE I: Input Parameter Extraction**

***Step 1:*** Extract the parameters $n$ and $N$ from the RC comprising of n lines $B_0, B_1, ..., B_{n-1}$ and N levels $G_0, G_1, ..., G_{N-1}$. Set fault-type and threshold value where 100 implies maximum fault coverage. Compute $max$-the maximum number of input test vectors for the RC.

$$n = 3 \quad N = 6 \quad max = 2^n = 8 \quad Threshold = 100$$
$$Input_Fault_Type = complete_missing_gate$$

***Step 2:*** Initialize $generation$, which indicates the current iteration, as 0.

**Algorithm 1** GA for Fault Detection in an RC

1: **INPUT** (RC, $Threshold$, $Input_Fault_Type$)
2: **OUTPUT** (fault coverage%, test set)
3: **use** (RC) **get** ($n$, $N$)
4: **procedure** SELECTION($n$) ▷ To get initial vector set
5: $max \leftarrow 2^n$
6: **for** $i \leftarrow 1, n$ **do**
7: $InitPop(V_i) \leftarrow$ **rand**(0,max)
8: **end for**
9: **end procedure**
10: **procedure** FITNESS($InitPop$)
11: **for** $i \leftarrow 1, n$ **do**
12: **use**($Input_Fault_Type$, $InitPop(V_i)$)
13: **get**($FaultCoverage$,$DetectedFaults$)
14: $FitnessFunction(V_i) \leftarrow FaultCoverage$
15: **if** $FitnessFunction(V_i) \geq Threshold$ **then**
16: Display Output, Exit
17: **end if**
18: **end for**
19: **end procedure**
20: **procedure** MULTIPLY($InitPop$,$FitnessFunction$)
21: **for** $i \leftarrow 1, n$ **do**
22: $Parent(V_{i1}, V_{i2}) \leftarrow$**Roulette**($InitPop$,$FitnessFunction$)
23: $ChildPopulation(V_i) \leftarrow$**CrossOver**($Parent(V_{i1}, V_{i2})$)
24: $ChildPopulation(V_i) \leftarrow$**Mutate**($Child(V_i)Population$)
25: **use Fitness**($ChildPopulation(V_i)$)
26: **get** $FitnessFunction(V_{i+n}), DetectedFaults$
27: **end for**
28: **end procedure**
29: $FinPop \leftarrow InitPop \bigcup ChildPopulation$
30: $FinFitness \leftarrow FitnessFunction$
31: **ArrangeDescending**($FinFitness$,$FinPop$)
32: **procedure** COMBINE($FinPop$,$DetectedFaults$)
33: **for** $i \leftarrow [\frac{n}{2}], |FinPop|$ **do**
34: $high$ = 0
35: **repeat**
36: $Combination_i(.) \leftarrow$**Combinations**(i,$FinPop$) ▷ combination taking i vectors
37: $fitness$ = **Fitness**($Combination(.)$)
38: **if** $fitness \geq Threshold$ **then**
39: Display Output, Exit
40: **end if**
41: **if** $high < fitness$ **then**
42: $high = fitness$
43: **end if**
44: **until** $high > fitness$ for $|FinPop|$ iterations
45: **end for**
46: $InitPop \leftarrow$ **MaxFitness**($Combination(.)$)
47: **end procedure**
48: Goto line 10

**STAGE II: Test Vector Selection**

***Step 3:*** Randomly select $n-$which is the upper limit on

0278-0070 (c) 2016 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCAD.2017.2695881, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems



the number of test vectors for $InitPop-$vectors from 0 to $(max-1)$-(rand(0,$max$)), both inclusive ensuring that none of the selected vectors repeat. $InitPop$ is a set containing all the initially selected $n$ vectors. $FinPop$ has elements copied from $InitPop$.
$InitPop = \{4, 6, 1\}$

**STAGE III: Fitness Function Computation**

***Step 4:*** Represent test vector in binary form, as in Fig. 7. Compute $FaultCoverage$ for CMGF as explained in Section. III-A. Here, test vector 4-[1 0 0], the first element in $InitPop$ is selected. The fault-free output $R$ and the faulty outputs after inducing the fault at each level are obtained.

$$R = 100$$
$$FaultyOutput = \{010, 010, 100, 100, 100, 100\}$$

***Step 5:*** Obtain $DetectedFault$ matrix.
if( $FaultyOutput(V_i)$ != $R$ )
set $DetectedFault_{[vector]}(V_i)$ to indicate detection
else
reset $DetectedFault_{[vector]}(V_i)$
Once $FaultyOutput$ has been completely traversed, count the number of 1's in the set $DetectedFault_{[vector]}$ and store it in $detected$. Find $FaultCoverage$ using Equation(1).
if( $FitnessFunction(V_i) \geq Threshold$ )
Goto Step 18.

$$DetectedFault_{[100]} = \{1, 1, 0, 0, 0, 0\}$$
$$CumulatedFault = n, \text{ see Section. III-A}$$
$$FaultCoverage = 33.33\%$$
$$FitnessFunction(V_i) = FaultCoverage = 33.33$$

***Step 6:*** Repeat steps 4 and 5 for all the vectors in $InitPop$. Copy members of $FitnessFunction$ into $FinFitness$.

$$DetectedFault_{[100]} = \{1, 1, 0, 0, 0, 0\}$$
$$DetectedFault_{[110]} = \{1, 1, 0, 0, 1, 1\}$$
$$DetectedFault_{[001]} = \{1, 0, 0, 0, 0, 0\}$$
$$FitnessFunction = \{33.33, 66.67, 16.67\}$$
$$FinFitness = \{33.33, 66.67, 16.67\}$$

**STAGE IV: Roulette Wheel**

***Step 7:*** Select pairs of vectors from $InitPop$. The process here follows the usual flow where vectors after being conformed onto a pie chart are picked two at a time based on their fitness functions. Select $n$ such pairs of parent.
$Parent = \{(100, 110), (110, 001), (110, 100)\}$

**STAGE V: Reproduction And Cross-Over**

***Step 8:*** A pair of parents is chosen and a random cross-over point is selected in the bit sequence. For 3_17$tc$, [1 0 0] and [1 1 0] are chosen as parents.
$CrossOverPoint = rand(0, n) = 1$
The child is generated taking bits before $CrossOverPoint$ from the first parent- '1' from [1 0 0]- and the remaining from the second parent- '1 0' from [1 1 0]- to yield [1 1 0] as the offspring. Continue until $n$ daughters have been generated.
$ChildPopulation = \{110, 111, 110\}$

**STAGE VI: Mutation**

***Step 9:*** In order to carry out mutation, a number is randomly selected from 0 and $(100^{generation+1} - 1)$, both inclusive. If this number is less than 10, then the test vector undergoes mutation where a random bit is selected and flipped. By doing this, the probability of mutation decreases in exponents of 10 for every $generation$. The genes after mutation are:
$ChildPopulation = \{110, 111, 100\}$

***Step 10:*** Repeated vectors are discarded and fitness function for the non-repeated ones is computed. The new members are added into the set $FinPop$ with their fitness values in $FinFitness$ along with faults detected by them in $DetectedFault$. The fitness values of the progeny are checked to see if $Threshold$ has been met.

$$FinPop = \{100, 110, 001, 111\}$$
$$FinFitness = \{33.33, 66.67, 16.67, 50\}$$
$$DetectedFault_{[111]} = \{1, 1, 1, 0, 0, 0\}$$

**STAGE VII: Test Population Generation**

***Step 11:*** Vectors of $FinPop$ are arranged in descending order based on their fitness values from $FinFitness$. Variable $m$ is defined as shown.

$$FinPop = \{110, 111, 100, 001\}$$
$$FinFitness = \{66.67, 50, 33.33, 16.67\}$$
$$m = |FinPop| = 4$$

**STAGE VIII: Minimal Test Set Generation From Test Population**

***Step 12:*** if( $\frac{n}{2} < 2$ ), $L$ = 2, else, $L$ = [$\frac{n}{2}$], rounded to the closest integral value. Combinations of $L$ vectors from $FinPop$ are made.

$$Combination_{21} = \{110, 111\}$$
$$Combination_{22} = \{110, 100\}$$
$$Combination_{23} = \{110, 001\}$$
$$Combination_{24} = \{111, 100\}$$
$$Combination_{25} = \{111, 001\}$$
$$Combination_{26} = \{100, 001\}$$

***Step 13:*** Set $upperBound$ as $m$. Reset $bestCoverage$ which is the best coverage that the algorithm can yield for the current test run.

***Step 14:*** Initialize $x$ and $i$ as 0. Reset $high$ which gives the best fitness value for the current combination. Update $DetectedFault$ and $detected$ for each of the $Combination_{Li}$ and compute $Fitness$.
if( $Fitness \geq Threshold$ )
$FinalTestSet = Combination_{Li}$, goto Step 18.
if( $Fitness > high$ )

0278-0070 (c) 2016 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCAD.2017.2695881, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems



$high$=$Fitness$
else, Increment $x$
if( $bestCoverage < high$ )
$bestCoverage$ = $high$, $FinalTestSet$=$Combination_{Li}$
if( $x = upperBound$ )
Goto Step 15
Increment $i$ and repeat Step 14 until all combinations of vectors are made or if x hits $upperBound$. For 3_17$tc$ with combinations of 2 vectors,
$Fitness = \{83.33, 66.67, 66.67, 50\}$
$high = 83.33$ $x$ = 4
***Step 15:*** if( $L = n$ ), Goto Step 16, else, Increment $L$ and repeat Step 14.

$$Combination_{31} = \{110, 111, 100\}$$
$$Combination_{32} = \{110, 111, 001\}$$
$$Combination_{33} = \{110, 100, 001\}$$
$$Combination_{34} = \{111, 100, 001\}$$
$$Fitness = \{83.3, 83.33, 66.67, 50\}$$
$$high = 83.33\% \qquad bestCoverage = 83.33\%$$

**STAGE IX: Initial Population for filial generation**
***Step 16:*** if( $|FinalTestSet| \leq n - 1$ )
Copy all vectors from $FinalTestSet$ into $InitPop$ and fill up the remaining slots upto $n$ using random selection method described in Step 3.
else, feed $(n - 1)$ vectors from $FinalTestSet$ into $InitPop$ and select 1 vector randomly using random selection method described in Step 3 and add it to $InitPop$.
$InitPop = \{110, 111, 101\}$
$FinPop = \{110, 111, 101\}$
Vector [1 0 1] is randomly generated.
$generation$ is incremented.
***Step 17:*** Goto Step 4 and repeat the process for the next generation. The values obtained are:
$ChildPopulation = \{101, 101, 100\}$
$FinPop = \{101, 110, 111, 100\}$
$FinFitness = \{83.33, 66.67, 50, 33.33\}$
For the combination, $Combination_{21} = \{101, 110\}$ the fitness is 100. Since $Threshold$ is met, goto Step 18.
$DetectedFault_{[110]||[101]} = \{1, 1, 1, 1, 1, 1\}$

**STAGE X: Optimum Test Set**
***Step 18:*** Display the test set and fault coverage.
$FinalTestSet = \{101, 110\}$, $bestCoverage = 100$, Exit.

Since this approach is heuristic, maximum coverage for every test run cannot be expected. In case the algorithm runs indefinitely with no change in $bestCoverage$, the process is exited prematurely and restarted.

### C. GA incorporating Directed search

In GA with Directed search, the initial population is selected based on the configuration of the first gate in the RC. Values for $InitPop$ are selected such that the output of the target is 1, which is ensured by forcing opposite values to the AND of control inputs and the target. Directed search makes use of the fact that by assigning opposite values to the control and target of the first gate, maximum faults can be detected. In case of Toffoli, for instance, whose function is given as (A, B, C) = (A, B, A·B⊕C), line C value toggles if A·B ≠ C. When A·B = 1, and C is 0, faults: Sa0-A, Sa0-B, Sa1-C, CMGF, PMGF-A, PMGF-B, BF-A·C, BF- B·C, BF-A+C and BF-B+C a total of 10 faults can be detected. Whereas, with test vector [0 1 0], 7 faults and with [1 1 1], 6 faults can be detected. By selecting a test vector which can detect

**Algorithm 2** Directed Search for $k$-CNOT Circuits

1: **INPUT** (RC)
2: **OUTPUT** ($InitPop$)
3: **use**(RC)**get**($controls$-$\{B_{c0}, B_{c1}...B_{ck}\}$,$target$ − $\{B_t\}$ of first gate with $B_0...B_{n-1}$ as input lines;$n$)
4: **procedure** SELECTION($n, B_t, B_{c0}, B_{c1}, ..., B_{ck}$)
5: m←1
6: **if** $|controls|$ = $\emptyset$ **then**
7: Goto line 30
8: **end if**
9: $v \leftarrow |target|$ +$|controls|$
10: $B_t \leftarrow 0$
11: $B_{c0}, B_{c1}, ..., B_{ck}$ ←11...1 ▷ set all control bits
12: **for** $i \leftarrow 0, n{-}1$ **do**
13: **if** $B_i \notin target$ **and** $B_i \notin controls$ **then**
14: $B_i \leftarrow$ **rand**(0,2) ▷ random allocation of 0/1 for remaining input lines
15: **end if**
16: **end for**
17: $InitPop(V_m)$ ←**Decimal**($B_{n-1}B_{n-2}...B_0$)
18: increment $m$
19: $B_t \leftarrow 1$
20: $var$ ←00...0 ▷ $var$, a ($k$+1) bit binary variable is reset
21: **for** $j \leftarrow 1, 2^{v-1} - 1$ **do**
22: **if** $m \leq n{-}2$ **then**
23: $B_{c0}B_{c1}...B_{ck} \leftarrow var$
24: increment $var$
25: repeat lines 12−18
26: **else**
27: Exit **for**
28: **end if**
29: **end for**
30: **for** $m \leftarrow m, n - 2$ **do**
31: $InitPop(V_m) \leftarrow$ **rand**(0,max)
32: **end for**
33: $InitPop(V_m)$ ←**Decimal**(111...11)
34: increment $m$
35: $InitPop(V_m)$ ←**Decimal**(000...00)
36: **end procedure**

0278-0070 (c) 2016 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCAD.2017.2695881, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems



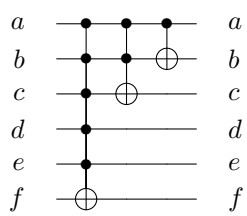


Fig. 8: A 3 level RC with multiple control $k$-CNOT gates

a majority of faults, probability of fault propagation increases, and with it, faster convergence of the algorithm is possible. Test set generation for GA incorporating Directed search, is explained in Algorithm 2 for $k$-CNOT circuits. The example circuit as shown in Fig. 8 is used to explain the method. It consists of 3 gates with the first gate comprising of 5 controls and a target $f$.

The values for the $InitPop$ are selected from $\{00000, 00001, 00010, ..., 11110\}$. The target is always opposite in value to the AND$ed$ output of all the controls thereby, ensuring target output as 1 [32]. A vector with all bits set as 1 or 0 detect a large number of faults especially SaF in most circuits [48]. Hence, $InitPop = \{111110, 000001, 000011, 000101, 000000, 111111\}$.

### *D. Illustration of proposed algorithm using GA with Directed search for Complete missing-gate fault detection*

The fault coverage and the test set for the example circuit in Fig. 6 is obtained by substituting Directed search for Random search in Illustration 2.

**Illustration 2** Illustration for GA using Directed Search for the RC 3_17$tc$

**INPUT:** RC with 'n' lines and 'N' gates; $Threshold$
**OUTPUT:** The test set and fault coverage

**STAGE I: Input Parameter Extraction**

Extracting and initializing the appropriate values.
$n = 3 \quad N = 6 \quad Threshold = 100$
$Input_Fault_Type = complete_missing_gate$

**STAGE II: Test Vector Selection**

Using the method explained in Algorithm 2, the initial population is selected.

$$InitPop = \{5, 7, 0\} \quad FinPop = \{5, 7, 0\}$$

**STAGE III: Fitness Function Computation**

Fitness function for the selected vectors and the $DetectedFaults$ are determined.

$$DetectedFault_{[101]} = \{1, 1, 1, 1, 0, 1\}$$
$$DetectedFault_{[111]} = \{1, 1, 1, 0, 0, 0\}$$
$$DetectedFault_{[000]} = \{1, 0, 1, 1, 1, 1\}$$
$$FitnessFunction = \{83.33, 50, 83.33\}$$

**STAGE IV: Roulette Wheel**

Parent population is arrived at using roulette wheel.

$$Parent = \{(000, 111), (000, 101), (101, 111)\}$$

**STAGE V: Reproduction And Cross-Over**

The parent chromosomes undergo reproduction to produce off-springs.

$$ChildPopulation = \{000, 000, 101\}$$

**STAGE VI: Mutation**

Mutation is carried out and $FinPop$ is arrived at. For 3_17$tc$, mutation did not occur for any of the vectors.

$$FinPop = \{101, 111, 000\}$$
$$FinFitness = \{83.33, 50, 83.33\}$$

Since fitness value did not meet $Threshold$, move to the next stage.

**STAGE VII: Test Population Generation**

Vectors of $FinPop$ are arranged in descending order based on their corresponding fitness values in $FinFitness$.

$$FinPop = \{101, 000, 111\}$$
$$FinFitness = \{83.33, 83.33, 50\} \; m = |FinPop| = 3$$

**STAGE VIII: Minimal Test Set Production From Test Population**

Combinations of vectors are made and checked for fitness function.
$Combination_{21} = \{101, 000\}, \quad Fitness = \{100\}$
Threshold has been met. Goto Stage X.

**STAGE X: Output**

$FinalTestSet = \{101, 000\}, \quad bestCoverage = 100$
Exit.

TABLE II: Comparison of GA with Directed and Random search methods for detecting CMGF in the circuit 3_17$tc^{\#}$

| Parameter | Directed | Random | †Directed |
|---|---|---|---|
| Time(avg)⋆ | <0.01 | <0.01 | <0.01 |
| V(avg)⋆ | 2 | 2 | 2 |
| $generation$(avg)⋆ | 1 | 1.467 | 1 |
| $P$(100% Fault Coverage) | 1 | 0.8 | 1 |

#Values are for 30 test runs ⋆Values are for 100% fault coverage †For Mirrored RC

A comparison of the results obtained using GA with Directed search and Random search for the circuit in Fig. 6 is presented in Table II for CMGF.

Directed search approach poses a problem for circuits with constant inputs. In such cases, only certain vectors would be permissible which are selected and added into $InitPop$. In case the upper limit has not been met, vectors are randomly selected and added to the set as in the case of GA with Random search. The algorithm described here is extended to the other fault models as well. The only difference arises in the computation of

0278-0070 (c) 2016 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCAD.2017.2695881, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems



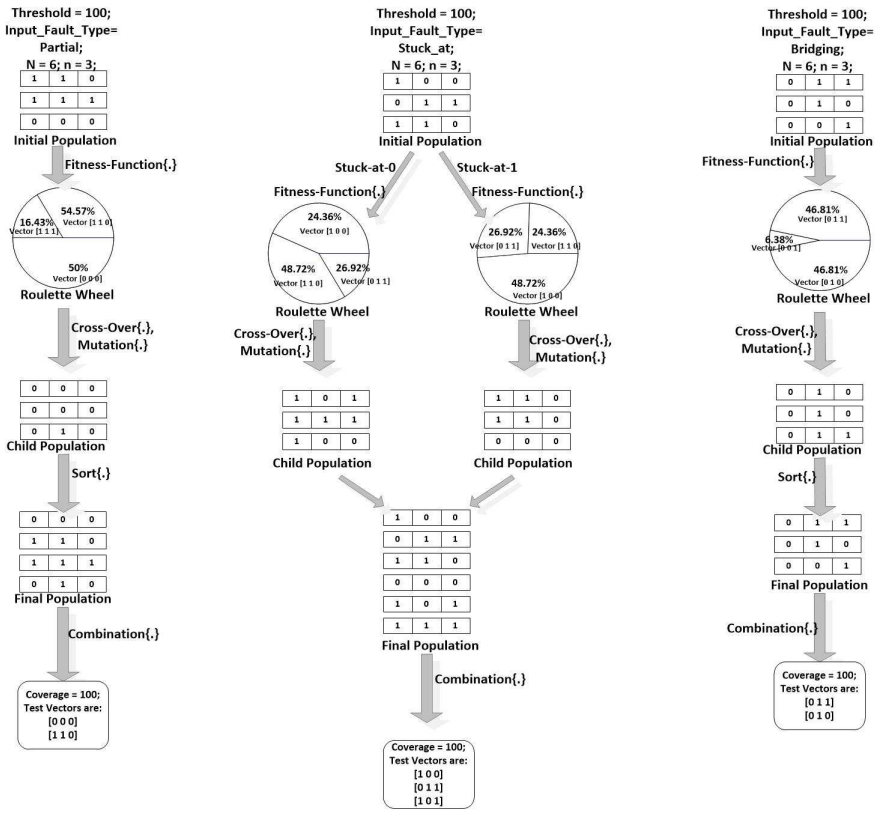


Fig. 9: GA applied to detect (a) PMGF, (b) SaF and (c) BF in 3_17$tc$

fault coverage% which is carried out as described in Section III-A.

In the case of SaF, through analysis, it was found that fault coverage was always 50% when both Sa0 and Sa1 faults are considered. Since fitness function remains constant, the selection of members in $ParentPop$ cannot be directly adopted. This problem is addressed by considering the fault coverage for Sa0 and Sa1 separately. Stages III through VI of the algorithm is performed separately for Sa0 and Sa1. $CumulatedFault$ would be half the original value and the faults introduced would be either a 0 or a 1. The $ChildPopulation$ obtained at the end of the two processes are combined and added into $FinPop$. Because of the branching, $|FinPop|$ generally doubles. $DetectedFault$ is found anew by introducing both Sa0 and Sa1 type of faults for all the members of the set $FinPop$. The rest of the process follows as in PMGF and CMGF. The process for PMGF, SaF and BF is depicted in Fig. 9.

The methodology developed in this section can be applied to the mirrored RC with data traversing from the last gate to the first gate. While doing so, the garbage outputs must take on appropriate constant values to preserve the reversible properties of the former. In Table II results of mirrored 3_17$tc$ have been added to illustrate the applicability of the algorithm when an input is fed to the last gate. Directed search is performed by feeding opposite values to the inputs $b$ and $c$ for the inverted circuit. One of the possible sets of $InitPop$ would be $\{000, 111, 010\}$.

Directed search has been devised to cater to the requirements of $k$-CNOT circuits, as, most benchmark circuits commence with a $k$-CNOT gate. The functionality of Fredkin gate requires its control to be activated for swapping contents of signal values on its target lines. In order to obtain maximum coverage, controls must be activated and targets must take on opposite values. RCs with Toffoli and Fredkin with Toffoli as the first gate can be Directed using the same approach. It can be observed from Section V that, there has been a significant improvement for $k$-CNOT and Peres based circuits using Directed approach and a slight improvement for Fredkin based circuits.

### *E. Test set generation for the set of all missing-gate Faults*

The algorithm developed here is extended to obtain a single test set for the set of all missing-gate faults. Both

0278-0070 (c) 2016 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCAD.2017.2695881, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems



CMGF and PMGF are induced individually on the RC as discussed in Section III-A in Stage III of the Algorithm 1 by following the remaining steps without any change. In this case, the $DetectedFault$ matrix takes on a larger dimension with the combined faults and results in a single test set for missing-gate faults.

## V. Results and Discussions

In this section, results obtained using the two approaches of GA are presented along with comparisons with methods proposed in the recent literature. Circuit information for every benchmark circuit is provided with $n$ indicating the number of lines, $N$ the number of gates and superscripts on every benchmark name to indicate the type of gate library, that the circuit has been implemented from. The circuits are sorted based on $n$. Computation time stated are processor specific. Tables V − VI carry out a comparison of the values obtained using GA with other proposed methods. All the existing works compared here consider 100% fault coverage. GA being a heuristic approach, cannot guarantee global minima in all cases like exact method. As is to be expected, the test set cardinality using GA has increased in a few cases over exact which can be observed in Tables V and VI. GA still remains a viable option as exact cannot be used without expending a great deal of effort for medium and large circuits. Table VIII gives the values for GA with Directed and Random search approach for each of the faults as discussed in Section IV. It can be observed that the values obtained using Directed approach have surpassed values obtained using the Random approach in terms of time taken, coverage and number of test vectors, in most cases. As the complexity of the circuit decreases, the values obtained using the two methods nearly coincide which can be attributed to the smaller $|InitPop|$. In the case of symmetric and cyclic circuits, as bit width increases, the disparity in the results obtained using the two methods is very large with Directed search being more favorable. This can be observed in $cycle17_3$ circuit for PMGF. These results lead to the conclusion that GA being a powerful tool can be amplified in terms of effectiveness by directing the search in the initial stage. Table VII shows the values obtained when the algorithm is extended to produce a single test set for missing-gate fault detection.

### A. Comparison of results for BF

Table V compares results obtained for BF with those proposed in [35] and [49]. There is an average reduction of 37.14% test vectors with GA compared to [49]. For the same test set, time taken by GA is less than in exact approach although the test set cardinality has increased by an average of 27%.

TABLE III: Comparison of CMGF detection parameters

| Ref | Benchmark Circuit | $N$ | $n$ | Ref. Paper | | Directed | | | Random | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | #V | time(s) | #FC % | #V | time(s) | #FC % | #V | time(s) |
| [13] | $ham7tc^{T}$ | 23 | 7 | 4 | 6.93 | 100 | 4 | 0.046 | 100 | 5 | 0.375 |
| | $mod5adders^{T}$ | 21 | 6 | 3 | 1.056 | 100 | 3 | 0.015 | 100 | 3 | 0.265 |
| | $mod5d1^{T}$ | 8 | 5 | 1 | 0.0479 | 100 | 1 | <0.01 | 100 | 1 | <0.01 |
| | $hwb4tc^{T}$ | 17 | 4 | 2 | 0.058 | 100 | 2 | <0.01 | 100 | 3 | 0.109 |
| | $rd32^{T}$ | 4 | 4 | 2 | 0.0202 | 100 | 2 | <0.01 | 100 | 3 | <0.01 |
| | $3_17tc^{T}$ | 6 | 3 | 2 | 0.017 | 100 | 2 | <0.01 | 100 | 2 | 0.015 |
| [11] | $hwb5_13^{T}$ | 88 | 28 | 8 | 0.05 | 100 | 5 | 0.772 | 100 | 6 | 0.797 |
| | $rd84_142^{T}$ | 28 | 15 | 9 | 0.06 | 100 | 6 | 0.046 | 100 | 4 | 0.359 |
| | $sym9_148^{T}$ | 210 | 10 | 32 | 0.59 | 100 | 1 | 0.29 | 100 | 1 | 0.297 |
| | $4gt12-v0_86^{T}$ | 14 | 5 | 3 | <0.01 | 100 | 3 | 0.082 | 100 | 2 | <0.01 |
| | $one-two-three-v0_97^{T}$ | 11 | 5 | 3 | <0.01 | 100 | 3 | 0.015 | 100 | 3 | <0.01 |
| [12] | $hwb5_13$ | 88 | 28 | 7 | 0.24 | 100 | 5 | 0.578 | 100 | 5 | 0.563 |
| | $rd84_142^{T}$ | 28 | 15 | 8 | 0.14 | 100 | 5 | 0.046 | 100 | 9 | 0.125 |
| | $sym9_148^{T}$ | 210 | 10 | 14 | 1.79 | 100 | 1 | 0.525 | 100 | 5 | 24.456 |
| | $4gt12-v0_86^{T}$ | 14 | 5 | 5 | 0.02 | 100 | 2 | <0.01 | 100 | 2 | 0.015 |
| | $one-two-three-v0_97^{T}$ | 11 | 5 | 3 | 0.02 | 100 | 3 | ¡0.01 | 100 | 3 | 0.015 |

#V = No. Of Test Vectors for detection #FC % = Fault Coverage Percentage $x^{T}$ = Toffoli based circuit

TABLE IV: Comparison of PMGF detection parameters

| Ref | Benchmark Circuit | $N$ | $n$ | Ref. Paper | | Directed | | | Random | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | #V | time(s) | #FC % | #V | time(s) | #FC % | #V | time(s) |
| [13] | $ham7tc^{T}$ | 23 | 7 | 4 | 6.83 | 100 | 4 | 0.062 | 100 | 4 | 0.328 |
| | $mod5adders^{T}$ | 21 | 6 | 3 | 1.076 | 100 | 3 | 0.015 | 100 | 3 | 0.078 |
| | $mod5d1^{T}$ | 8 | 5 | 1 | 0.0486 | 100 | 1 | < 0.01 | 100 | 2 | 0.062 |
| | $hwb4tc^{T}$ | 17 | 4 | 2 | 0.054 | 100 | 2 | 0.015 | 100 | 2 | 0.062 |
| | $rd32^{T}$ | 4 | 4 | 2 | 0.0196 | 100 | 2 | 0.015 | 100 | 2 | 0.062 |
| | $3_17tc^{T}$ | 6 | 3 | 2 | 0.016 | 100 | 2 | <0.01 | 100 | 2 | 0.015 |
| [11] | $hwb5_13^{T}$ | 88 | 28 | 9 | 0.07 | 100 | 5 | 0.578 | 100 | 5 | 0.563 |
| | $rd84_142^{T}$ | 28 | 15 | 120 | 0.03 | 100 | 5 | 0.046 | 100 | 9 | 0.125 |
| | $sym9_148^{T}$ | 210 | 10 | 120 | 3.71 | 100 | 1 | 0.525 | 100 | 5 | 24.456 |
| | $4gt12-v0_86^{T}$ | 14 | 5 | 10 | <0.01 | 100 | 2 | <0.01 | 100 | 2 | 0.015 |
| | $one-two-three-v0_97^{T}$ | 11 | 5 | 3 | 0.02 | 100 | 3 | ¡0.01 | 100 | 3 | 0.015 |

#V = No. Of Test Vectors for detection #FC % = Fault Coverage Percentage $x^{T}$ = Toffoli based circuit

TABLE V: Comparison of BF Detection Parameters

| Ref | Benchmark Circuit | $N$ | $n$ | Ref. Paper | | Directed | | | Random | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | #V | time(s) | #FC % | #V | time(s) | #FC % | #V | time(s) |
| [49] | $rd32^{T}$ | 28 | 15 | 6 | $\star$ | 100 | 4 | <0.01 | 100 | 3 | 0.015 |
| | $ham7tc^{T}$ | 23 | 7 | 4 | $\star$ | 100 | 6 | 0.343 | 100 | 6 | 0.187 |
| | $graycode6^{T}$ | 5 | 6 | 9 | $\star$ | 100 | 3 | 0.031 | 100 | 5 | 0.093 |
| | $xor5d1^{T}$ | 4 | 5 | 8 | $\star$ | 100 | 3 | 0.015 | 100 | 4 | 0.062 |
| | $4_49-12-32^{T}$ | 12 | 4 | 5 | $\star$ | 100 | 4 | 0.015 | 100 | 4 | <0.01 |
| | $ham3tc^{T}$ | 5 | 3 | 3 | $\star$ | 100 | 3 | <0.01 | 100 | 3 | 0.015 |
| [35] | $gf2\hat{}4mult_19_83^{T}$ | 19 | 12 | 4 | 76.3734 | 98.56 | 8 | 1.093 | 98.64 | 6 | 0.656 |
| | $nth_prime6_inc_55_667^{T}$ | 55 | 6 | 4 | 116.3606 | 100 | 6 | 3.219 | 100 | 6 | 1.673 |
| | $2of5d1^{T}$ | 18 | 6 | 4 | 25.4031 | 100 | 5 | 0.343 | 100 | 5 | 0.031 |
| | $5mod5_fc^{T}$ | 10 | 6 | 3 | 4.768 | 100 | 4 | 0.109 | 100 | 5 | 0.015 |

#V = No. Of Test Vectors for detection #FC % = Fault Coverage Percentage $x^{T}$ = Toffoli based circuit $\star$time not mentioned in reference [49]

TABLE VI: Comparison of SaF Detection Parameters with [13]

| Benchmark Circuit | $N$ | $n$ | Ref. Paper | | Directed | | | Random | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | #V | time(s) | #FC % | #V | time(s) | #FC % | #V | time(s) |
| $ham7tc^{T}$ | 23 | 7 | 3 | 7.126 | 100 | 5 | 0.078 | 100 | 5 | 0.187 |
| $mod5adders^{T}$ | 21 | 6 | 3 | 1.12 | 100 | 3 | 0.046 | 100 | 4 | 0.265 |
| $mod5d1^{T}$ | 8 | 5 | 3 | 0.1866 | 98.89 | 4 | 0.031 | 98.89 | 4 | 0.171 |
| $hwb4tc^{T}$ | 17 | 4 | 3 | 0.0839 | 100 | 3 | <0.01 | 100 | 3 | 0.046 |
| $rd32^{T}$ | 4 | 4 | 3 | 0.0408 | 97.5 | 3 | <0.01 | 97.5 | 4 | 0.125 |
| $3_17tc^{T}$ | 6 | 3 | 3 | 0.018 | 100 | 3 | <0.01 | 100 | 3 | 0.031 |

#V = No. Of Test Vectors for detection #FC % = Fault Coverage Percentage $x^{T}$ = Toffoli based circuit

0278-0070 (c) 2016 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCAD.2017.2695881, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems



TABLE VII: Test set and coverage for combination of missing-gate faults- CMGF and PMGF

| Benchmark Circuit | $N$ | $n$ | Faults | Directed | | | Random | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | #V | TIME(s) | #FC % | #V | time(s) | #FC % |
| $gf2$^$10mult$_109_509$^{T}$ | 109 | 30 | 318 | 2 | 4.142 | 100 | 20 | 276.76 | 71.70 |
| $cycle17_3^{T}$ | 48 | 20 | 457 | 3 | 2.531 | 100 | 10 | 32.111 | 1.31 |
| $cnt3-5_180^{T}$ | 20 | 16 | 60 | 1 | 0.657 | 100 | 2 | 0.719 | 100 |
| 0410184_169$^{T}$ | 46 | 14 | 95 | 4 | 0.109 | 100 | 9 | 0.218 | 100 |
| $sym6_316^{T}$ | 29 | 14 | 72 | 11 | 0.062 | 100 | 10 | 0.078 | 100 |
| $nth_prime7_inc$_1427_3172$^{T}$ | 1427 | 7 | 3197 | 7 | 706.275 | 98 | 7 | 735.567 | 96.90 |

#V = No. Of Test Vectors for detection #FC % = Fault Coverage Percentage $x^{T}$ = Toffoli based circuit

### B. Comparison of results for missing-gate faults

Tables III and IV compare the values obtained for missing-gate type faults with exact method as described in [13], SAT method in [11] and PBO in [12]. The number of test vectors required for complete fault coverage in the case of GA is found to be the same in most of the cases as that using the exact approach with lesser computation time. Test set requirement for full coverage is much less compared to SAT and PBO based approaches with an average decrease of 72.73% and 56.76% test vectors, respectively. PMGF results compared with [11] show a decrease of 93.89% test vectors on average. Directed search produced better results than Random search in a majority of circuits.

### C. Comparison of results for SaF

Table VI compares values obtained for SaF type faults using GA and exact approach stated in [13]. GA takes much lesser time for computation in all cases with nearly the same number of test vectors.

### D. Comparison of GA incorporating Directed and Random approach

Figs. 10 through 13 show the variation of fault coverage percentage with $generation$. Directed approach selects individuals which generally have higher fitness function than most vectors in the gene pool while in Random search approach, the probability of selection of the same vector would be $\frac{1}{2^n}$ which is a very small value as bit size increases. In the case of $ham15-109-214$, though the initial selection was poorer in GA with the Directed approach than in the Random approach, it produced maximum coverage in 2 $generation$ while random approach required 4 due to constancy in fault coverage% for 3 such generations. Circuits $gf2$^$10mult$_109_509, $sym9$_148 and $cycle10$_2 converge within one generation for Directed approach. The Random approach in $gf2$^$10mult$_109_509 yields a fault coverage% less than 65% even with 25 $generation$ while Directed approach produces the result in a single $generation$ demonstrating the superiority of Directed approach over Random when applied to $k$-CNOT circuits with high input variables.

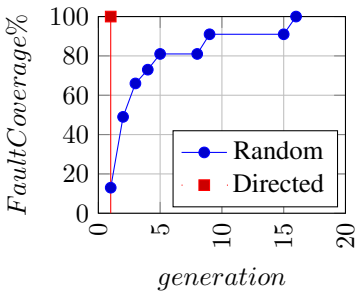


Fig. 10: Variation of fault coverage with $generation$ for the circuit $cycle10_2$ using GA with Random and Directed approaches

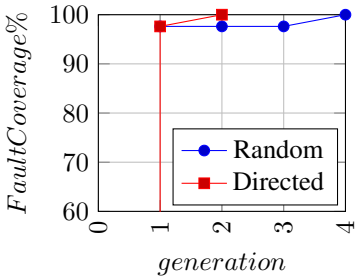


Fig. 11: Variation of fault coverage with $generation$ for the circuit $ham15-109-214$ using GA with Random and Directed approaches

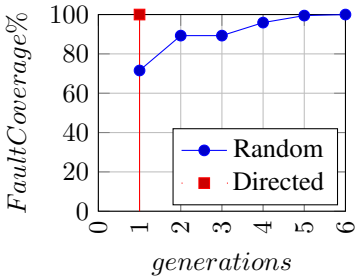


Fig. 12: Variation of fault coverage with $generation$ for the circuit $sym9_148$ using GA with Random and Directed approaches

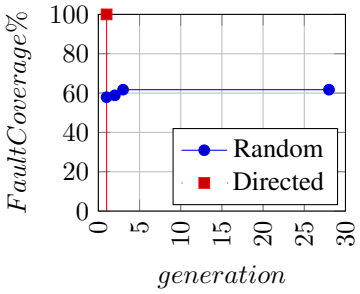


Fig. 13: Variation of fault coverage with $generation$ for the circuit $gf2$^$10mult$_109_509 using GA with Random and Directed approaches

0278-0070 (c) 2016 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCAD.2017.2695881, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems



TABLE VIII: Test set and coverage for PMGF, CMGF, SaF, BF using Directed and Random search approach for benchmark circuits

| | | | PMGF | | | | | | | CMGF | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Directed | | | Random | | | | Directed | | | Random | | |
| BenchMark Circuit | $N$ | $n$ | Total Faults | #FC % | #V | time(s) | #FC % | #V | time(s) | Total Faults | #FC % | #V | time(s) | #FC % | #V | time(s) |
| $bw_291^{T}$ | 307 | 87 | 307 | 100 | 4 | 9.593 | 100 | 5 | 16.406 | 432 | 100 | 4 | 3.5 | 100 | 5 | 6.421 |
| $hwb7_302^{F}$ | 281 | 73 | 281 | 100 | 26 | 39.85 | 100 | 29 | 39.96 | 426 | 100 | 17 | 31.09 | 100 | 18 | 33.67 |
| $hwb6_14^{F}$ | 159 | 46 | 159 | 100 | 9 | 5.76 | 100 | 9 | 6.64 | 241 | 100 | 11 | 4.328 | 100 | 11 | 4.69 |
| $ham15_298^{T}$ | 153 | 45 | 153 | 100 | 1 | 0.109 | 100 | 2 | 4.48 | 157 | 100 | 1 | 0.09 | 100 | 2 | 4.142 |
| $cycle10_293^{T}$ | 78 | 39 | 78 | 100 | 2 | 0.828 | 100 | 3 | 1.203 | 98 | 100 | 1 | 0.09 | 100 | 2 | 6.143 |
| $rd84_313^{T}$ | 104 | 34 | 104 | 100 | 16 | 1.31 | 100 | 21 | 2.734 | 143 | 100 | 19 | 1.33 | 100 | 23 | 2.69 |
| $gf2\hat{}10mult_109_509^{T}$ | 109 | 30 | 209 | 100 | 2 | 2.546 | 61.72 | 4 | 74.039 | 109 | 100 | 2 | 1.828 | 63.3 | 15 | 24.845 |
| $t-add-8^{T}$ | 122 | 24 | 160 | 100 | 14 | 1.339 | 88.13 | 22 | 6.437 | 122 | 100 | 14 | 1.132 | 90.16 | 17 | 4.296 |
| $mux_246^{T}$ | 35 | 22 | 131 | 25.95 | 2 | 0.156 | 4.58 | 2 | 5.54 | 35 | 97.14 | 22 | 12.156 | 34.29 | 17 | 2.063 |
| $cycle17_3^{T}$ | 48 | 20 | 409 | 100 | 3 | 2.071 | 0.73 | 2 | 39.799 | 48 | 100 | 3 | 0.908 | 6.25 | 2 | 9.015 |
| $ryy6_256^{T}$ | 44 | 17 | 328 | 37.8 | 8 | 10.859 | 37.8 | 8 | 15.562 | 44 | 100 | 2 | 1.671 | 54.55 | 2 | 6.656 |
| $cnt3-5_180^{T}$ | 20 | 16 | 40 | 100 | 2 | 0.078 | 100 | 2 | 0.453 | 20 | 100 | 2 | 0.594 | 100 | 2 | 0.328 |
| $ham15tc1^{T}$ | 132 | 15 | 352 | 95.74 | 15 | 12.178 | 93.18 | 8 | 45.348 | 132 | 96.97 | 14 | 4.851 | 98.48 | 9 | 24.455 |
| $0410184_169^{T}$ | 46 | 14 | 49 | 100 | 4 | 0.078 | 100 | 3 | 0.062 | 46 | 100 | 5 | 0.062 | 100 | 4 | 0.063 |
| $sym6_316^{T}$ | 29 | 14 | 43 | 100 | 11 | 0.062 | 100 | 10 | 0.031 | 29 | 100 | 10 | 0.062 | 100 | 11 | 0.031 |
| $squar5_261^{T}$ | 43 | 13 | 95 | 100 | 7 | 0.265 | 100 | 6 | 0.219 | 43 | 100 | 8 | 0.281 | 100 | 9 | 0.203 |
| $cycle10_2^{T}$ | 19 | 12 | 100 | 100 | 2 | 0.062 | 100 | 2 | 0.07 | 19 | 100 | 2 | 0.062 | 100 | 2 | 0.201 |
| $sym9_148^{T}$ | 210 | 10 | 756 | 100 | 1 | 0.525 | 100 | 1 | 17.273 | 210 | 100 | 1 | 0.29 | 100 | 1 | 9.592 |
| $sys6-v0_144^{T+P}$ | 15 | 10 | 23 | 100 | 4 | 0.015 | 100 | 4 | 0.015 | 15 | 100 | 3 | 0.015 | 100 | 3 | 0.078 |
| $urf_2_277^{T}$ | 3144 | 8 | 3777 | 94.07 | 8 | 74.455 | 94.07 | 8 | 74.455 | 44 | 100 | 2 | 1.671 | 54.55 | 2 | 6.656 |
| $nth_prime7_inc_1427_3172^{T}$ | 1427 | 7 | 1770 | 95.58 | 7 | 16.957 | 95.99 | 7 | 16.304 | 1427 | 97.83 | 7 | 14.909 | 96.71 | 7 | 43.472 |
| $hwb7_60^{F}$ | 166 | 7 | 396 | 100 | 7 | 9.375 | 100 | 7 | 8.937 | 30 | 100 | 7 | 13.329 | 98.80 | 7 | 49.23 |
| $rd53_131^{T}$ | 28 | 7 | 24 | 100 | 5 | 0.015 | 100 | 6 | 0.046 | 28 | 100 | 7 | 0.062 | 100 | 6 | <0.01 |
| $hwb6_57^{F}$ | 65 | 6 | 122 | 100 | 5 | 0.438 | 100 | 5 | 1.515 | 65 | 100 | 6 | 2.249 | 100 | 6 | 1.218 |
| $hwb5_54^{F}$ | 24 | 5 | 30 | 100 | 4 | 0.015 | 100 | 4 | 0.125 | 24 | 100 | 4 | 0.0310 | 100 | 5 | 0.421 |
| $4gt12-v0_86^{T}$ | 14 | 5 | 20 | 100 | 3 | 0.015 | 100 | 3 | 0.015 | 14 | 100 | 3 | 0.11 | 100 | 2 | <0.01 |
| $one-two-three-v0_97^{T}$ | 11 | 5 | 23 | 100 | 3 | <0.01 | 100 | 3 | <0.01 | 11 | 100 | 3 | 0.015 | 100 | 3 | 0.015 |
| $alu-v0_26^{T}$ | 6 | 5 | 8 | 100 | 2 | 0.015 | 100 | 2 | <0.01 | 6 | 100 | 2 | 0.015 | 100 | 2 | 0.015 |
| $hwb4_51^{F}$ | 11 | 4 | 10 | 100 | 2 | 0.015 | 100 | 2 | 0.015 | 11 | 100 | 2 | 0.015 | 100 | 2 | 0.015 |
| | | | SaF | | | | | | | BF | | | | | | |
| | | | | Directed | | | Random | | | | Directed | | | Random | | |
| BenchMark Circuit | $N$ | $n$ | Total Faults | #FC % | #V | time(s) | #FC % | #V | time(s) | Total Faults | #FC % | #V | time(s) | #FC % | #V | time(s) |
| $bw_291^{T}$ | 307 | 87 | 53592 | 82.28 | 5 | 663.81 | - | - | †TO | 2304456 | - | - | †TO | - | - | †TO |
| $hwb7_302^{F}$ | 281 | 73 | 41172 | 99.9 | 7 | 1971.547 | - | - | †TO | 1482192 | - | - | †TO | - | - | †TO |
| $hwb6_14^{F}$ | 159 | 46 | 14720 | 82.29 | 6 | 79.296 | 82.29 | 6 | 84.95 | 331200 | 87.13 | 6 | 681.655 | 86.54 | 14 | 2075.418 |
| $ham15_298^{T}$ | 153 | 45 | 13860 | 80.84 | 5 | 527.958 | 80.84 | 6 | 975.797 | 304920 | 91.26 | 9 | 5057.34 | 87 | 4 | 4649.65 |
| $cycle10_293^{T}$ | 78 | 39 | 6162 | 83.14 | 19 | 109.056 | 82.88 | 19 | 140.925 | 117078 | 92.12 | 11 | 680.173 | 91.84 | 4 | 791.142 |
| $rd84_313^{T}$ | 104 | 34 | 7140 | 79.43 | 8 | 252.99 | 79.3 | 8 | 235.418 | 117810 | 87.55 | 22 | 677.573 | 87.43 | 22 | 627.14 |
| $gf2\hat{}10mult_109_509^{T}$ | 109 | 30 | 6600 | 95.5 | 7 | 121.809 | 88.83 | 15 | 102.88 | 95700 | 97.16 | 15 | 577.157 | 92.68 | 15 | 718.27 |
| $t-add-8^{T}$ | 122 | 24 | 5904 | 91.82 | 12 | 46.877 | 89.33 | 12 | 46.877 | 67896 | 97.46 | 14 | 259.306 | 95.81 | 13 | 263.918 |
| $mux_246^{T}$ | 35 | 22 | 1584 | 99.94 | 11 | 3.687 | 86.36 | 11 | 3.437 | 16632 | 100 | 14 | 90.024 | 93.94 | 11 | 14.534 |
| $cycle17_3^{T}$ | 48 | 20 | 1960 | 100 | 7 | 34.769 | 90.2 | 10 | 16.174 | 18620 | 99.33 | 13 | 81.332 | 97.79 | 10 | 143.758 |
| $ryy6_256^{T}$ | 44 | 17 | 1530 | 99.87 | 12 | 73.207 | 96.01 | 18 | 10.515 | 12240 | 99.98 | 9 | 31.316 | 99.77 | 8 | 61.864 |
| $cnt3-5_180^{T}$ | 20 | 16 | 672 | 92.56 | 8 | 9.438 | 92.56 | 8 | 18.297 | 5040 | 97.34 | 15 | 3.093 | 97.62 | 9 | 31.051 |
| $ham15tc1^{T}$ | 132 | 15 | 3990 | 100 | 10 | 63.631 | 100 | 7 | 30.126 | 27930 | 100 | 12 | 151.116 | 100 | 9 | 96.02 |
| $0410184_169^{T}$ | 46 | 14 | 1316 | 100 | 6 | 1.687 | 100 | 6 | 1.578 | 8554 | 100 | 11 | 4.484 | 100 | 8 | 4.172 |
| $sym6_316^{T}$ | 29 | 14 | 840 | 91.31 | 7 | 0.891 29 | 91.31 | 7 | 0.922 | 5460 | 96.15 | 12 | 1.687 | 96.08 | 13 | 13.609 |
| $squar5_261^{T}$ | 43 | 13 | 1144 | 94.67 | 6 | 2.82 | 94.67 | 9 | 2.658 | 6864 | 95.92 | 8 | 8.628 | 95.92 | 7 | 4.047 |
| $cycle10_2^{T}$ | 19 | 12 | 480 | 100 | 3 | 0.875 | 100 | 4 | 0.546 | 2640 | 100 | 7 | 3.046 | 100 | 7 | 2.391 |
| $sys6-v0_144^{T+P}$ | 15 | 10 | 320 | 95.63 | 5 | 0.141 | 95.63 | 7 | 0.515 | 1350 | 98.47 | 7 | 0.25 | 98.47 | 8 | 0.829 |
| $sym9_148^{T}$ | 210 | 10 | 4220 | 99.98 | 10 | 101.898 | 99.27 | 10 | 93.485 | 18990 | 100 | 9 | 43.89 | 100 | 9 | 48.831 |
| $urf_2_277^{T}$ | 3144 | 8 | 50320 | 99.9 | 8 | 184.11 | 99.86 | 8 | 723.68 | 176064 | 99.73 | 8 | 2590.541 | 99.73 | 8 | 2597.485 |
| $nth_prime7_inc_1427_3172^{T}$ | 1427 | 7 | 19992 | 99.78 | 7 | 391.9 | 99.86 | 7 | 229.698 | 59976 | 99.7 | 7 | 693.734 | 99.7 | 7 | 701.58 |
| $hwb7_60^{F}$ | 166 | 7 | 2338 | 100 | 2 | 7.5 | 100 | 5 | 6.875 | 7014 | 100 | 7 | 520.108 | 99.86 | 7 | 7.344 |
| $rd53_131^{T}$ | 28 | 7 | 406 | 98.28 | 3 | 0.062 28 | 98.28 | 4 | 0.078 | 1218 | 99.84 | 5 | 0.124 | 99.84 | 5 | 0.234 |
| $hwb6_57^{F}$ | 65 | 6 | 792 | 100 | 2 | 0.641 | 100 | 5 | 0.593 | 1980 | 100 | 5 | 1.344 | 100 | 5 | 1.032 |
| $hwb5__54^{F}$ | 24 | 5 | 250 | 100 | 2 | 0.046 | 100 | 5 | 1.375 | 500 | 100 | 4 | 0.266 | 100 | 4 | 0.016 |
| $4gt12-v0_86^{T}$ | 14 | 5 | 150 | 95.33 | 4 | 0.062 | 95.33 | 4 | 0.015 | 300 | 100 | 4 | 0.015 | 100 | 4 | 0.031 |
| $one-two-three-v0_97^{T}$ | 11 | 5 | 120 | 95.83 | 4 | 1.171 | 95.83 | 3 | <0.01 | 240 | 86.67 | 4 | 0.015 | 86.67 | 3 | 0.01 |
| $alu-v0_26^{T}$ | 6 | 5 | 70 | 100 | 3 | <0.01 | 100 | 5 | 0.016 | 140 | 100 | 4 | <0.01 | 100 | 4 | <0.01 |
| $hwb4_51^{F}$ | 11 | 4 | 96 | 100 | 2 | 0.015 | 100 | 4 | <0.01 | 144 | 100 | 2 | 0.078 | 100 | 2 | <0.01 |

#V = No. Of Test Vectors for detection #FC % = Fault Coverage Percentage $x^{T}$ = Toffoli based circuit $x^{F}$ = Fredkin based circuit $x^{P}$ = Peres based circit †TO - time out

0278-0070 (c) 2016 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCAD.2017.2695881, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems



## VI. Conclusion

In this paper, a GA based platform is used to obtain a test set for optimum coverage using less computation time for small, medium and large circuits. This is the first attempt in reversible literature to use principles of GA for test set generation with optimum fault coverage for a wide class of faults. A GA based method for fault detection in RCs designed with $k$-CNOT, Peres and Fredkin gates has been proposed. An algorithm for a Directed search method exploring unique properties of reversible circuits as opposed to the random search approach inherent to GA is also presented. Significant improvement is observed with respect to the number of generations, computation time and fault coverage. It is observed that the algorithm worked very well with circuits having small, medium and large input lines in the case of missing-gate faults. Detection of SaF and BF in large RCs takes longer convergence time owing to the increased fault cardinality in the order of $10^6$. It would be interesting to combine circuit partitioning method with GA to catalyze convergence. The fault detection methodology used here can be extended to other subsets of reversible fault models such as Additional gate and Additional control faults.

To conclude, by using GA proposed in this paper teamed with an appropriate algorithm to direct the initial stage, faster convergence can be ensured. In RCs having few input lines but a large number of gates, the time taken for convergence was found to be relatively higher. Hence, future work would involve exploring ways to decrease the number of test vectors, reduce test generation complexity by using a suitable circuit partitioning technique and to devise an algorithm for new quantum fault models. Fault diagnosis using GA has also been earmarked for future research.

## References


[1] R. Landauer, "Irreversibility and heat generation in the computing process," *IBM journal of research and development*, vol. 5, no. 3, pp. 183–191, 1961.

[2] C. H. Bennett, "Logical reversibility of computation," *IBM journal of Research and Development*, vol. 17, no. 6, pp. 525–532, 1973.

[3] M. A. Nielsen and I. L. Chuang, *Quantum computation and quantum information*. Cambridge university press, 2010.

[4] C. Taraphdar, T. Chattopadhyay, and J. N. Roy, "Mach–zehnder interferometer-based all-optical reversible logic gate," *Optics & Laser Technology*, vol. 42, no. 2, pp. 249–259, 2010.

[5] X. Ma, J. Huang, C. Metra, and F. Lombardi, "Reversible gates and testability of one dimensional arrays of molecular qca," *Journal of Electronic Testing*, vol. 24, no. 1, pp. 297–311, 2008. [Online]. Available: http://dx.doi.org/10.1007/s10836-007-5042-2

[6] I. Polian, T. Fiehn, B. Becker, and J. P. Hayes, "A family of logical fault models for reversible circuits," in *14th Asian Test Symposium (ATS'05)*, Dec 2005, pp. 422–427.

[7] A. Mishchenko and M. Perkowski, "Logic synthesis of reversible wave cascades," Jun 2002.

[8] T. Toffoli, "Reversible computing," in *International Colloquium on Automata, Languages, and Programming*. Springer, 1980, pp. 632–644.

[9] E. Fredkin and T. Toffoli, "Conservative logic," in *Collision-based computing*. Springer, 2002, pp. 47–81.

[10] A. Peres, "Reversible logic and quantum computers," *Physical review A*, vol. 32, no. 6, p. 3266, 1985.

[11] H. Zhang, R. Wille, and R. Drechsler, "Sat-based atpg for reversible circuits," in *IEEE 5th International Design and Test Workshop*, Dec 2010, pp. 149–154.

[12] R. Wille, H. Zhang, and R. Drechsler, "Atpg for reversible circuits using simulation, boolean satisfiability, and pseudo boolean optimization," in *IEEE Computer Society Annual Symposium on VLSI*, July 2011, pp. 120–125.

[13] A. Nagamani, S. Ashwin, B. Abhishek, and V. Agrawal, "An exact approach for complete test set generation of toffoli-fredkin-peres based reversible circuits," *Journal of Electronic Testing*, vol. 32, no. 2, pp. 175–196, 2016.

[14] G. J. Woeginger, "Exact algorithms for np-hard problems: A survey," in *Combinatorial OptimizationEureka, You Shrink!* Springer, 2003, pp. 185–207.

[15] M. Srinivas and L. M. Patnaik, "A simulation-based test generation scheme using genetic algorithms," in *VLSI Design, Proceedings. IEEE The Sixth International Conference on*, Jan 1993, pp. 132–135.

[16] N. Alhagi, M. Hawash, and M. Perkowski, "Synthesis of reversible circuits with no ancilla bits for large reversible functions specified with bit equations," in *40th IEEE International Symposium on Multiple-Valued Logic*, May 2010, pp. 39–45.

[17] M. B. Tahoori, M. Momenzadeh, J. Huang, and F. Lombardi, "Defects and faults in quantum cellular automata at nano scale," in *VLSI Test Symposium, 2004. Proceedings. 22nd IEEE*. IEEE, 2004, pp. 291–296.

[18] D. Y. Feinstein, V. Nair, and M. A. Thornton, "Advances in quantum computing fault tolerance and testing." in *HASE*. Citeseer, 2007, pp. 369–370.

[19] M. Lukac, M. Kameyama, M. Perkowski, P. Kerntopf, and C. Moraga, "Fault models in reversible and quantum circuits," in *Advances in Unconventional Computing*. Springer, 2017, pp. 475–493.

[20] R. Wille, D. Große, L. Teuber, G. W. Dueck, and R. Drechsler, "RevLib: An online resource for reversible functions and reversible circuits," in *International Symposium on Multi-Valued Logic*, 2008, pp. 220–225.

[21] D. Maslov, G. Dueck, and N. Scott, "Reversible logic synthesis benchmarks page," *Online: http://www. cs. uvic. ca/˜ dmaslov*, 2005.

[22] D. P. Vasudevan, P. K. Lala, J. Di, and J. P. Parkerson, "Reversible-logic design with online testability," *IEEE Transactions on Instrumentation and Measurement*, vol. 55, no. 2, pp. 406–414, April 2006.

[23] S. N. Mahammad and K. Veezhinathan, "Constructing online testable circuits using reversible logic," *Instrumentation and Measurement, IEEE Transactions on*, vol. 59, no. 1, pp. 101–109, 2010.

[24] D. P. Vasudevan, P. K. Lala, and J. P. Parkerson, "Online testable reversible logic circuit design using nand blocks," in *Defect and Fault Tolerance in VLSI Systems, (DFT) Proceedings. 19th IEEE International Symposium on*. IEEE, 2004, pp. 324–331.

[25] N. Farazmand, M. Zamani, and M. B. Tahoori, "Online fault testing of reversible logic using dual rail coding," in *On-Line Testing Symposium (IOLTS), IEEE 16th International*. IEEE, 2010, pp. 204–205.

[26] B. Sen, J. Das, and B. K. Sikdar, "A dft methodology targeting online testing of reversible circuit," in *Devices, Circuits and Systems (ICDCS), International Conference on*, March 2012, pp. 689–693.

[27] K. N. Patel, J. P. Hayes, and I. L. Markov, "Fault testing for reversible circuits," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 23, no. 8, pp. 1220–1230, Aug 2004.


0278-0070 (c) 2016 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TCAD.2017.2695881, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems




[28] J. P. Hayes, I. Polian, and B. Becker, "Testing for missing-gate faults in reversible circuits," in *Test Symposium, 13th Asian*, Nov 2004, pp. 100–105.

[29] X. Fang-ying, C. Han-wu, L. Wen-jie, and L. Zhi-giang, "Fault detection for single and multiple missing-gate faults in reversible circuits," in *IEEE Congress on Evolutionary Computation*, June 2008, pp. 131–135.

[30] M. Zamani, N. Farazmand, and M. B. Tahoori, "Fault masking and diagnosis in reversible circuits," in *Test Symposium (ETS), 16th IEEE European*. IEEE, 2011, pp. 69–74.

[31] D. K. Kole, H. Rahaman, D. K. Das, and B. B. Bhattacharya, "Derivation of test set for detecting multiple missing-gate faults in reversible circuits," *Computers & Electrical Engineering*, vol. 39, no. 2, pp. 225–236, 2013.

[32] H. Rahaman, D. K. Kole, D. K. Das, and B. B. Bhattacharya, "On the detection of missing-gate faults in reversible circuits by a universal test set," in *21st International Conference on VLSI Design (VLSID)*. IEEE, 2008, pp. 163–168.

[33] J. Zhong and J. C. Muzio, "Analyzing fault models for reversible logic circuits," in *IEEE International Conference on Evolutionary Computation*, 2006, pp. 2422–2427.

[34] P. Sarkar and S. Chakrabarti, "Universal test set for bridging fault detection in reversible circuit," in *3rd International Design and Test Workshop*, Dec 2008, pp. 51–56.

[35] A. N. Nagamani, B. Abhishek, and V. K. Agrawal, "Deterministic approach for bridging fault detection in peres-fredkin and toffoli based reversible circuits," in *IEEE International Conference on Computational Intelligence and Computing Research (ICCIC)*, Dec 2015, pp. 1–6.

[36] M. Zamani and M. B. Tahoori, "Online missing/repeated gate faults detection in reversible circuits," in *Defect and Fault Tolerance in VLSI and Nanotechnology Systems (DFT), IEEE International Symposium on*. IEEE, 2011, pp. 435–442.

[37] A. Deb, D. K. Das, H. Rahaman, R. Wille, R. Drechsler, and B. B. Bhattacharya, "Reversible synthesis of symmetric functions with a simple regular structure and easy testability," *ACM Journal on Emerging Technologies in Computing Systems (JETC)*, vol. 12, no. 4, p. 34, 2016.

[38] G. Ascia, V. Catania, and M. Palesi, "A ga-based design space exploration framework for parameterized system-on-a-chip platforms," *IEEE Transactions on Evolutionary Computation*, vol. 8, no. 4, pp. 329–346, Aug 2004.

[39] C. Bernardeschi, L. Cassano, M. G. Cimino, and A. Domenici, "Gabes: A genetic algorithm based environment for seu testing in sram-fpgas," *Journal of Systems Architecture*, vol. 59, no. 10, pp. 1243–1254, 2013.

[40] F. Corno, P. Prinetto, M. Rebaudengo, and M. S. Reorda, "Gatto: a genetic algorithm for automatic test pattern generation for large synchronous sequential circuits," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 15, no. 8, pp. 991–1000, Aug 1996.

[41] D. G. Saab, Y. G. Saab, and J. A. Abraham, "Automatic test vector cultivation for sequential vlsi circuits using genetic algorithms," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 15, no. 10, pp. 1278–1285, Oct 1996.

[42] A. Paler, I. Polian, and J. P. Hayes, "Detection and diagnosis of faulty quantum circuits," in *17th Asia and South Pacific Design Automation Conference*, Jan 2012, pp. 181–186.

[43] I. Polian and J. P. Hayes, "Advanced modeling of faults in reversible circuits," in *Design Test Symposium (EWDTS), East-West*, Sept 2010, pp. 376–381.

[44] S. Sivanandam and S. Deepa, *Introduction to genetic algorithms*. Springer Science & Business Media, 2007.

[45] A. C. Koenig, "A study of mutation methods for evolutionary algorithms," *University of Missouri-Rolla*, 2002.

[46] K. Deb and D. Deb, "Analysing mutation schemes for real-parameter genetic algorithms," *International Journal of Artificial Intelligence and Soft Computing*, vol. 4, no. 1, pp. 1–28, 2014.

[47] K. Ramasamy, R. Tagare, E. Perkins, and M. Perkowski, "Fault localization in reversible circuits is easier than for classical circuits," 2004.

[48] H. Thapliyal, N. Ranganathan, and S. Kotiyal, "Design of testable reversible sequential circuits," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 21, no. 7, pp. 1201–1209, July 2013.

[49] H. Rahaman, D. K. K. D. K. Kole, D. K. Das, and B. B. Bhattacharya, "Optimum test set for bridging fault detection in reversible circuits," in *16th Asian Test Symposium (ATS)*, Oct 2007, pp. 125–128.

**Nagamani A N** received M.Tech degree from VTU, Belagavi, India and currently pursuing Ph.D from VTU, Belagavi in the area of Reversible circuit optimization and Testing. She is working with PES Institute of Technology as Assistant Professor in Department of Electronics and Communication Engineering. Her area of research includes, Reversible logic optimization and testing, low power Digital VLSI, Analog and mixed signal design, VLSI architecture for optimized performance. She has published several publications in the above mentioned areas in the peer reviewed conferences and journals.

**Anuktha S Nayak** is in the graduating class of Electronics and Communications engineering at PES Institute of Technology. Her current field of interests are Reversible Logic, Digital Communication and VLSI for Signal Processing.

**Nanditha N** is a graduate of Electronics and Communications engineering at PES Institute of Technology. Her current research interests focus on the areas of Reversible Logic, Memory Design and VLSI for Signal Processing.

**V. K. Agrawal** received the M.Tech. degree from IIT, Kanpur, and the Ph. D. degree from the Department of Computer Science, IISc, Bangalore, in 1986. He joined the ISRO Satellite Center in 1978. He has worked as a Group Director for the Control Systems Group at ISRO. His research interests are in the area of Emerging Technologies for circuit design and optimization. His expertise is in Petri nets and evolutionary computing. Presently, he is the Director for CORI (Crucible of Research and Innovation Lab) at PESIT, Bangalore, India. He has published several papers in peer reviewed conferences and journals in his specialized fields.

0278-0070 (c) 2016 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission. See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.